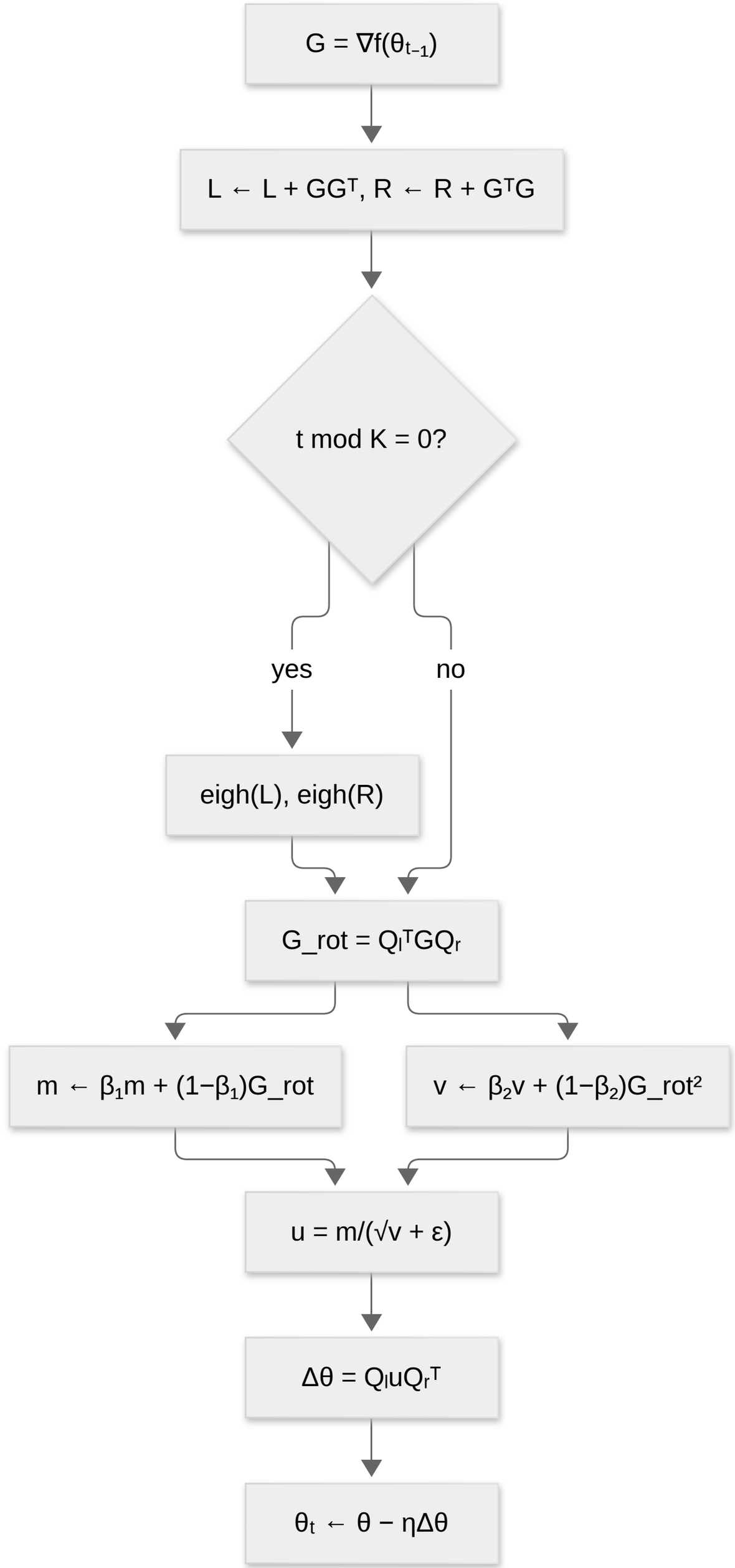

$G = \nabla f(\theta_{t-1})$
$L \leftarrow L + GG^T,\ R \leftarrow R + G^TG$
t mod K = 0?
yes
no
eigh(L), eigh(R)
$G_rot = Q_l^T G Q_r$
$m \leftarrow \beta_1 m + (1-\beta_1) G_rot$
$v \leftarrow \beta_2 v + (1-\beta_2) G_rot^2$
$u = m/(\sqrt{v} + \varepsilon)$
$\Delta\theta = Q_l u Q_r^T$
$\theta_t \leftarrow \theta - \eta\Delta\theta$